

Singular Value Decomposition Pt 2

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Rank of a Matrix: Row Rank

Definition: Row Rank

The **row rank** is the number of rows that cannot be written as a linear combination of other rows.

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 4 & 9 & 12 \end{bmatrix}$$

- Observe that: $\text{row } 3 = 2 \cdot \text{row } 1 + \text{row } 2$
- $\text{row rank}(\mathbf{A}) = 2$

Rank of a Matrix: Column Rank

Definition: Column rank

Column rank is the number of columns that cannot be written as a linear combination of other columns.

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 4 & 9 & 12 \end{bmatrix}$$

- ▶ Notice that: $\text{col}_3 = 3 \cdot \text{col}_1 + 0 \cdot \text{col}_2$
- ▶ $\text{col rank}(\mathbf{A}) = 2$

The Fundamental Property of Rank

$$\text{rank}(A) = \text{col rank}(A) = \text{row rank}(A)$$

Orthonormal Vectors

Definition

A set of vectors $v_1, v_2, \dots, v_k$ are **orthonormal** if they satisfy:

- **Orthogonal:** The vectors are perpendicular to each other.

$$v_i \cdot v_j = 0 \quad \text{for } i \neq j$$

- **Unit Length:** Each vector has a magnitude of 1.

$$v_i^T v_i = 1$$

Orthonormal Vectors

$$\begin{bmatrix} 2/3 \\ 2/3 \\ 1/3 \end{bmatrix}, \quad \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1/(3\sqrt{2}) \\ 1/(3\sqrt{2}) \\ -4/(3\sqrt{2}) \end{bmatrix}$$

Orthonormal Matrices

Consider the following matrix A :

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- ▶ **Rows of A :** The rows are orthonormal.
- ▶ **Columns of A :** The columns are orthonormal.
- ▶ **Orthogonal matrix**, meaning $A^T A = I$.

Singular Value Decomposition (SVD)

For an $m \times n$ matrix M , there exist matrices U, Σ, V such that:

$$M = U\Sigma V^T$$

- ▶ U : An $m \times r$ matrix.
- ▶ Σ : An $r \times r$ diagonal matrix of singular values.
- ▶ V : An $r \times n$ matrix.
- ▶ U, V : Matrices with orthonormal columns.

SVD Interpretation

The Singular Value Decomposition (SVD) of a matrix M factorizes it into three distinct matrices:

$$M = U\Sigma V^T$$

$$M = \underbrace{\begin{bmatrix} | & | & \\ \mathbf{u}_1 & \mathbf{u}_2 & \dots \\ | & | & \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \ddots \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ & \vdots & \end{bmatrix}}_{V^T}$$

Outer Product Expansion:

$$M = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

SVD interpretation

$$M = \underbrace{\sigma_1}_{\text{Wt of Concept 1}} \cdot \underbrace{\mathbf{u}_1}_{\text{Row Loads}} \cdot \underbrace{\mathbf{v}_1^T}_{\text{Column Loads}} + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$$

SVD interpretation

$$M = \begin{bmatrix} 5 & 5 & 4 & 1 & 1 \\ 1 & 3 & 1 & 5 & 5 \\ 4 & 4 & 3 & 0 & 0 \end{bmatrix}$$
$$= \begin{bmatrix} -.70 & .30 & -.64 \\ -.49 & -.86 & .12 \\ -.51 & .39 & .75 \end{bmatrix} \begin{bmatrix} 11.59 & & & & \\ & 6.33 & & & \\ & & 0.264 & & \end{bmatrix} \begin{bmatrix} -.53 & -.61 & -.42 & -.27 & -.27 \\ -.35 & .08 & .24 & -.63 & -.63 \\ -.19 & .72 & -.63 & -.13 & -.13 \end{bmatrix}$$

Rank k approximation

$$k = 2$$

$$M = \begin{bmatrix} 5 & 5 & 4 & 1 & 1 \\ 1 & 3 & 1 & 5 & 5 \\ 4 & 4 & 3 & 0 & 0 \end{bmatrix}$$
$$\approx \begin{bmatrix} -.70 & .30 \\ -.49 & -.86 \\ -.51 & .39 \end{bmatrix} \begin{bmatrix} 11.59 & 6.33 \end{bmatrix} \begin{bmatrix} -.53 & -.61 & -.42 & -.27 & -.27 \\ -.35 & .08 & .24 & -.63 & -.63 \end{bmatrix}$$

- ▶ u_i : rows involvements for concept i
 - ▶ $\gg 0$: strongly associated
 - ▶ $\ll 0$: strongly opposed
- ▶ v_i : cols involvements for concept i .
- ▶ σ_i : importance of concept i .

SVD and Big Data

- ▶ Interpret and explore big data (revealing hidden concepts)
- ▶ How to compute SVD efficiently?

Computing the SVD

Eigenvectors of a Matrix A :

A vector $\mathbf{v}$ is an eigenvector of A if it satisfies:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Why do we care about eigenvectors?

SVD: $M = U\Sigma V^T$.

- ▶ The column vectors of V are the **eigenvectors** of $M^T M$.
- ▶ The column vectors of U are the **eigenvectors** of MM^T .
- ▶ The non-zero singular values in Σ are the square roots of the eigenvalues of $M^T M$.

Computing Eigenvectors Efficiently

To find the largest eigenvalue λ_1 and its eigenvector $\mathbf{v}$, we use the **Power Iteration** method.

The Algorithm:

1. **Initialize:** Choose a starting vector $\mathbf{x}_0$ (often a vector of ones).

$$\mathbf{x}_0 = [1, 1, \dots, 1]^T$$

2. **Iterate:** For $k = 1, 2, 3, \dots$:

- ▶ Compute the next direction: $\mathbf{y} = A\mathbf{x}_k$
- ▶ **Normalize** to prevent the vector from growing to infinity:

$$\mathbf{x}_{k+1} = \frac{\mathbf{y}}{\|\mathbf{y}\|}$$

Result: As $k \rightarrow \infty$, the vector $\mathbf{y}_T$ converges to the eigenvector corresponding to the **largest eigenvalue** λ_1 .

Computing Subsequent Eigenvectors

$$A_{\text{new}} = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T$$

The Iterative Process:

- ▶ Apply the **Power Method** to A_{new} .
- ▶ The result will converge to the next eigenvector $\mathbf{v}_2$ and eigenvalue λ_2 .
- ▶ This process can be repeated ($A_{\text{next}} = A_{\text{new}} - \lambda_2 \mathbf{v}_2 \mathbf{v}_2^T$) to find $\mathbf{v}_3, \lambda_3$, and so on.